



Unit-2

Device management



agenda

Types of devices

Input/Output Devices Examples

Secondary storage
structure-Magnetic Disk
Structure

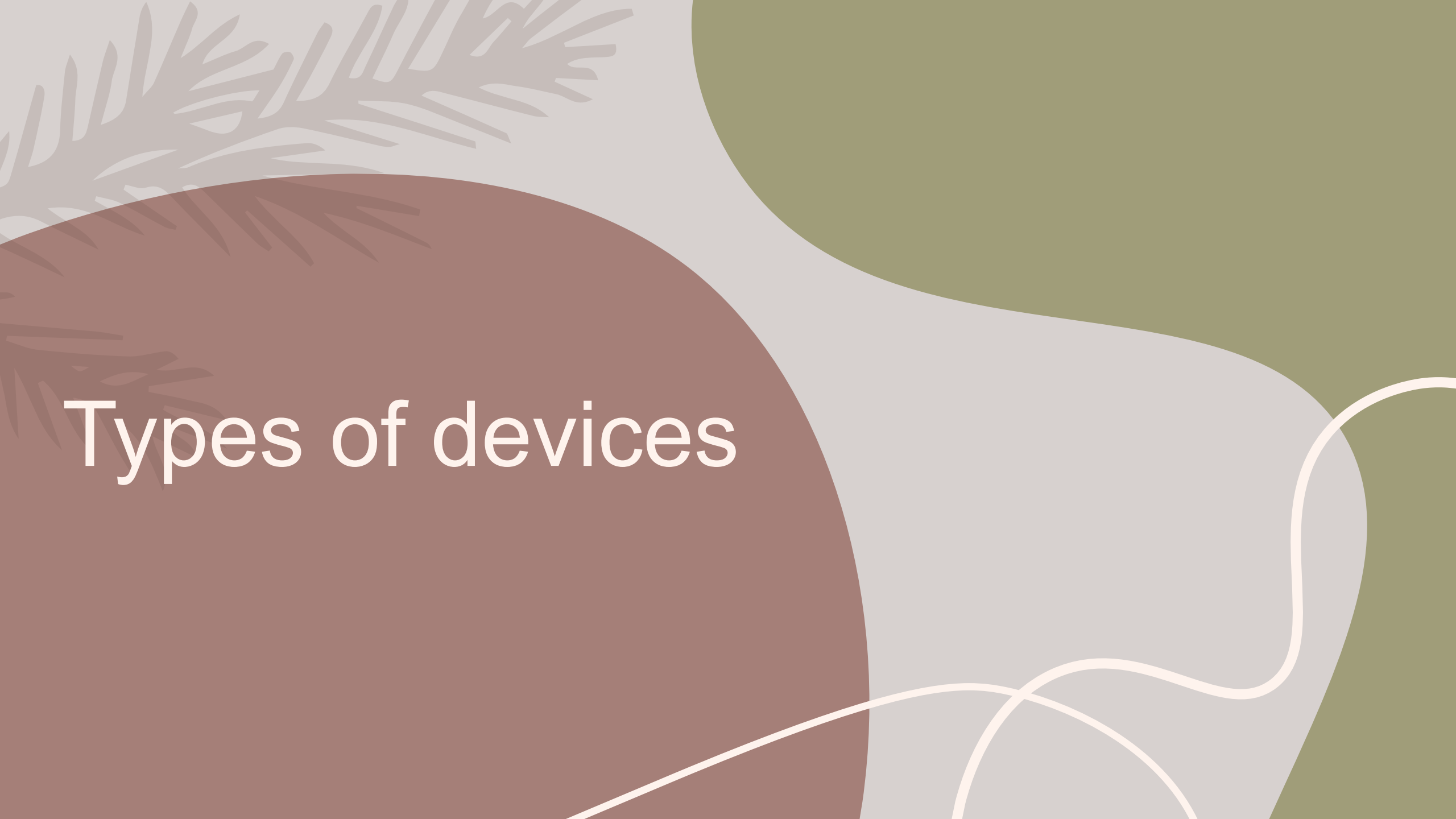
Disk Scheduling Algorithms

RAID

Device management

Device management in operating system known as the management of the I/O devices such as a keyboard, magnetic tape, disk, printer, microphone, USB ports, scanner, etc.as well as the supporting units like control channels.





Types of devices

Dedicated devices:-

- Such types of devices in the device management in an operating system are dedicated or assigned to only one job at a time until that job releases them.
- Devices like printers, tape drivers, plotters, etc. demand such an allocation scheme since it would be awkward if several users share them simultaneously.
- The disadvantage of such kind devices is inefficiency resulting from allocating the device to a single user for the entire duration of job execution even though the device is not put to use 100% of the time.

Shared devices:-

- These devices can be allocated to several processes. Disk-DASD can be shared among several processes at the same time by interleaving their requests.
- The interleaving is carefully controlled by the Device Manager and all issues must be resolved on the basis of predetermined policies.

Virtual Devices:-

- These devices are the combination of the first two types and they are dedicated devices which are transformed into shared devices. For example, a printer converted into a shareable device via spooling program which re-routes all the print requests to a disk.
- A print job is not sent straight to the printer, instead, it goes to the disk(spool) until it is fully prepared with all the necessary sequences and formatting, then it goes to the printers.
- This technique can transform one printer into several virtual printers which leads to better performance and use.

Input/Output Devices Examples

- Here are the list of some most popular and common input/output devices:
 - Keyboard
 - Mouse
 - Monitor
 - Modem
 - Scanner
 - Laser Printer
 - Ethernet
 - Diskis

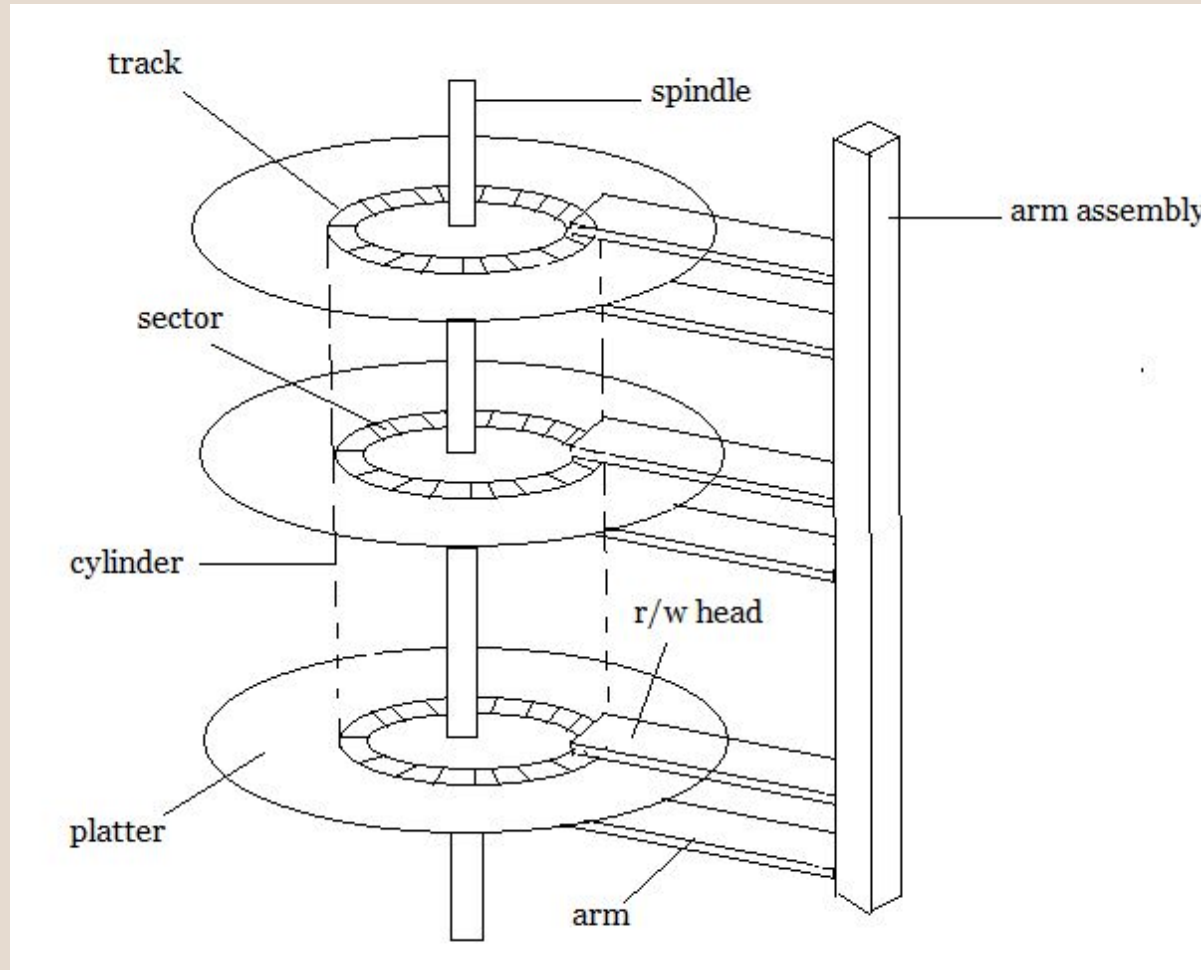
Storage devices

- There are two types of storage devices:-
 - Volatile Storage Device –
It loses its contents when the power of the device is removed.
 - Non-Volatile Storage device –
It does not lose its contents when the power is removed. It holds all the data when the power is removed.

Secondary storage structure

- Secondary storage devices are those devices whose memory is non volatile, meaning, the stored data will be intact even if the system is turned off. Here are a few things worth noting about secondary storage.
 - Secondary storage is also called auxiliary storage.
 - Secondary storage is less expensive when compared to primary memory like RAMs.
 - The speed of the secondary storage is also lesser than that of primary storage.
 - Hence, the data which is less frequently accessed is kept in the secondary storage.
 - A few examples are magnetic disks, magnetic tapes, removable thumb drives etc.

Magnetic Disk Structure



Structure of a magnetic disk

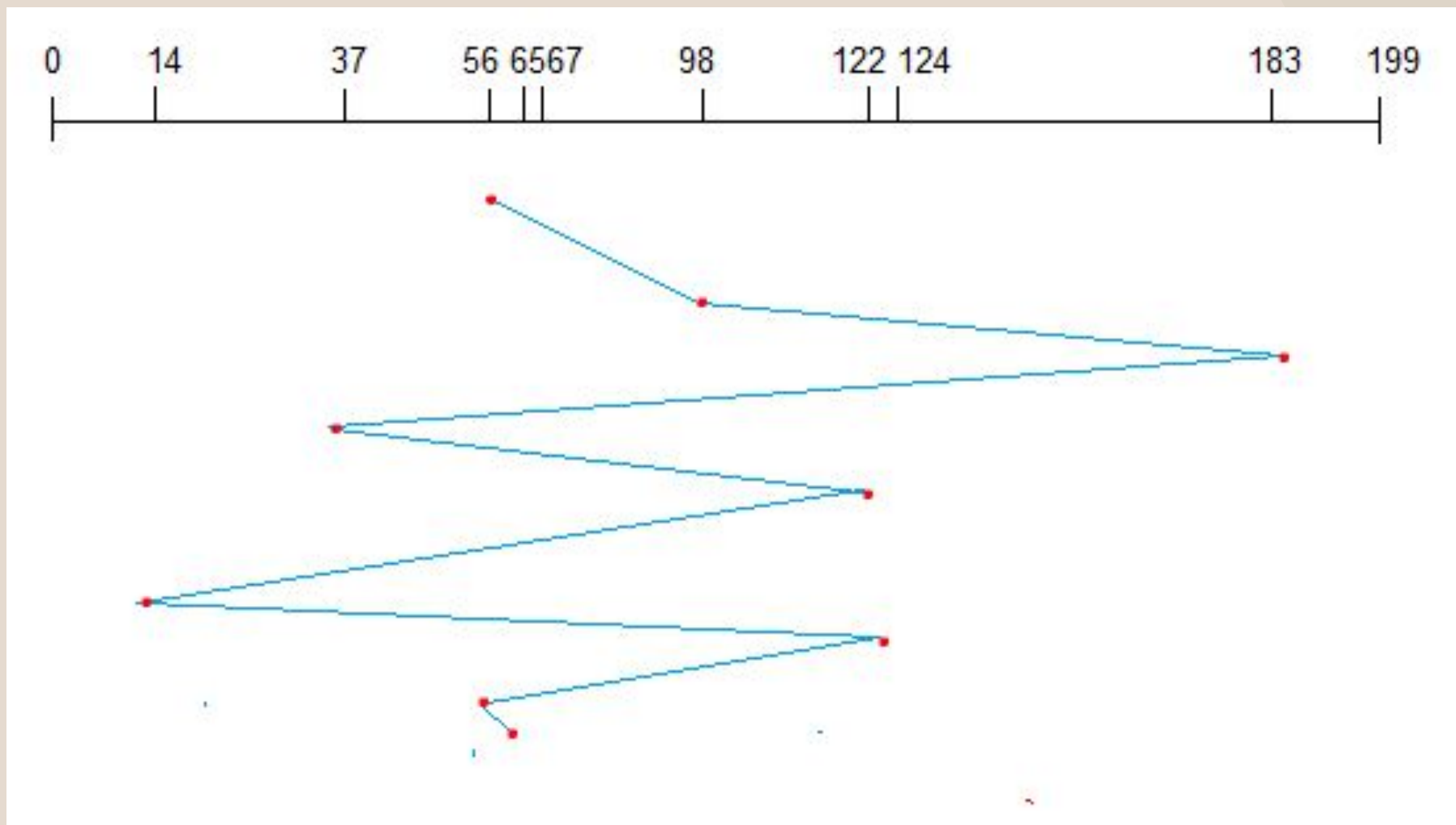
- A magnetic disk contains several platters. Each platter is divided into circular shaped tracks. The length of the tracks near the centre is less than the length of the tracks farther from the centre. Each track is further divided into sectors, as shown in the figure.
- Tracks of the same distance from centre form a cylinder. A read-write head is used to read data from a sector of the magnetic disk.

- The speed of the disk is measured as two parts:
 - Transfer rate: This is the rate at which the data moves from disk to the computer.
 - Random access time: It is the sum of the seek time and rotational latency.
- Seek time is the time taken by the arm to move to the required track.
- Rotational latency is defined as the time taken by the arm to reach the required sector in the track.

Disk Scheduling Algorithms

First Come First Serve

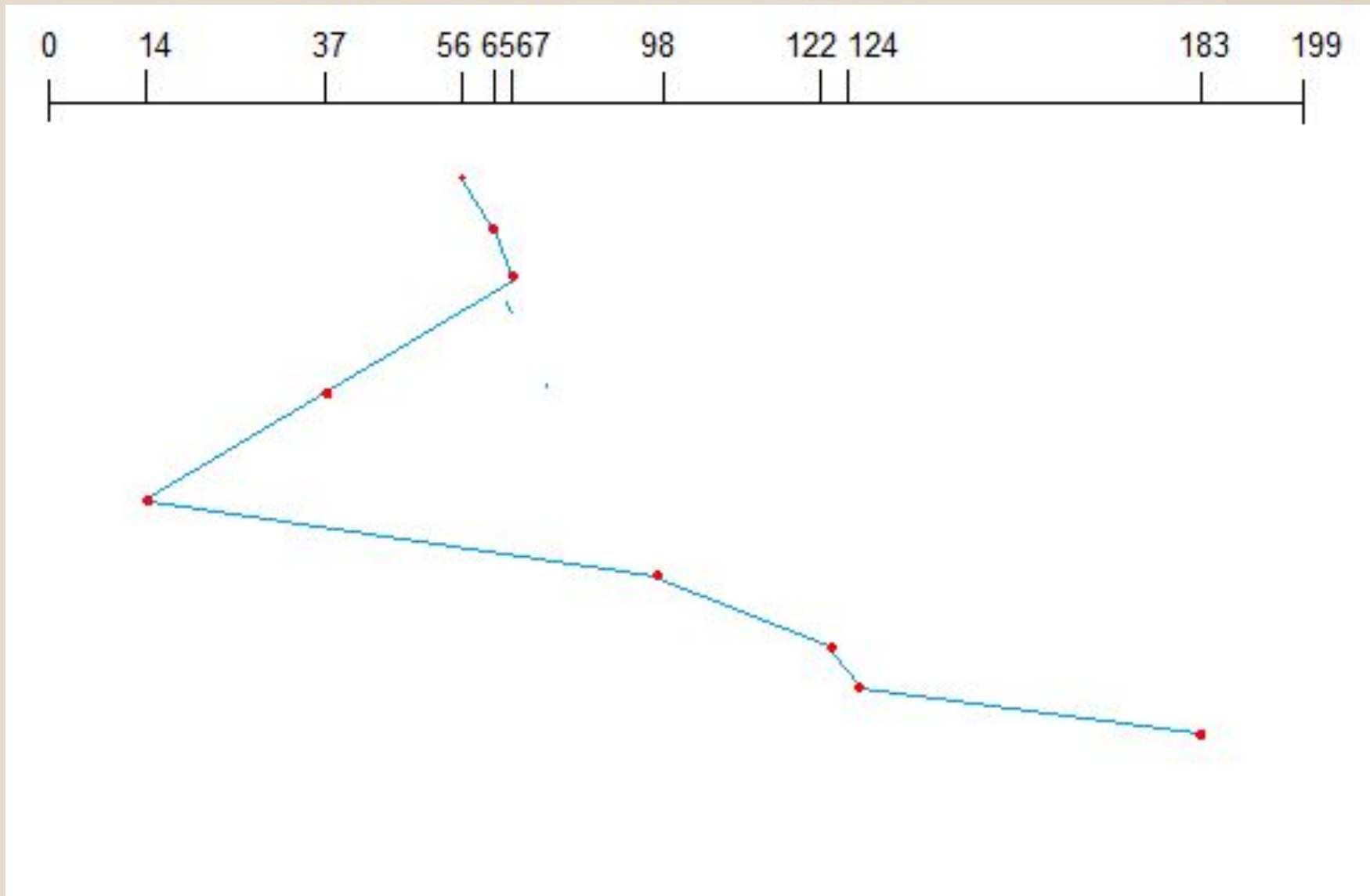
- This algorithm performs requests in the same order asked by the system. Let's take an example where the queue has the following requests with cylinder numbers as follows:
- 98, 183, 37, 122, 14, 124, 65, 67
- Assume the head is initially at cylinder 56. The head moves in the given order in the queue i.e., $56 \rightarrow 98 \rightarrow 183 \rightarrow \dots \rightarrow 67$.



Shortest Seek Time First (SSTF)

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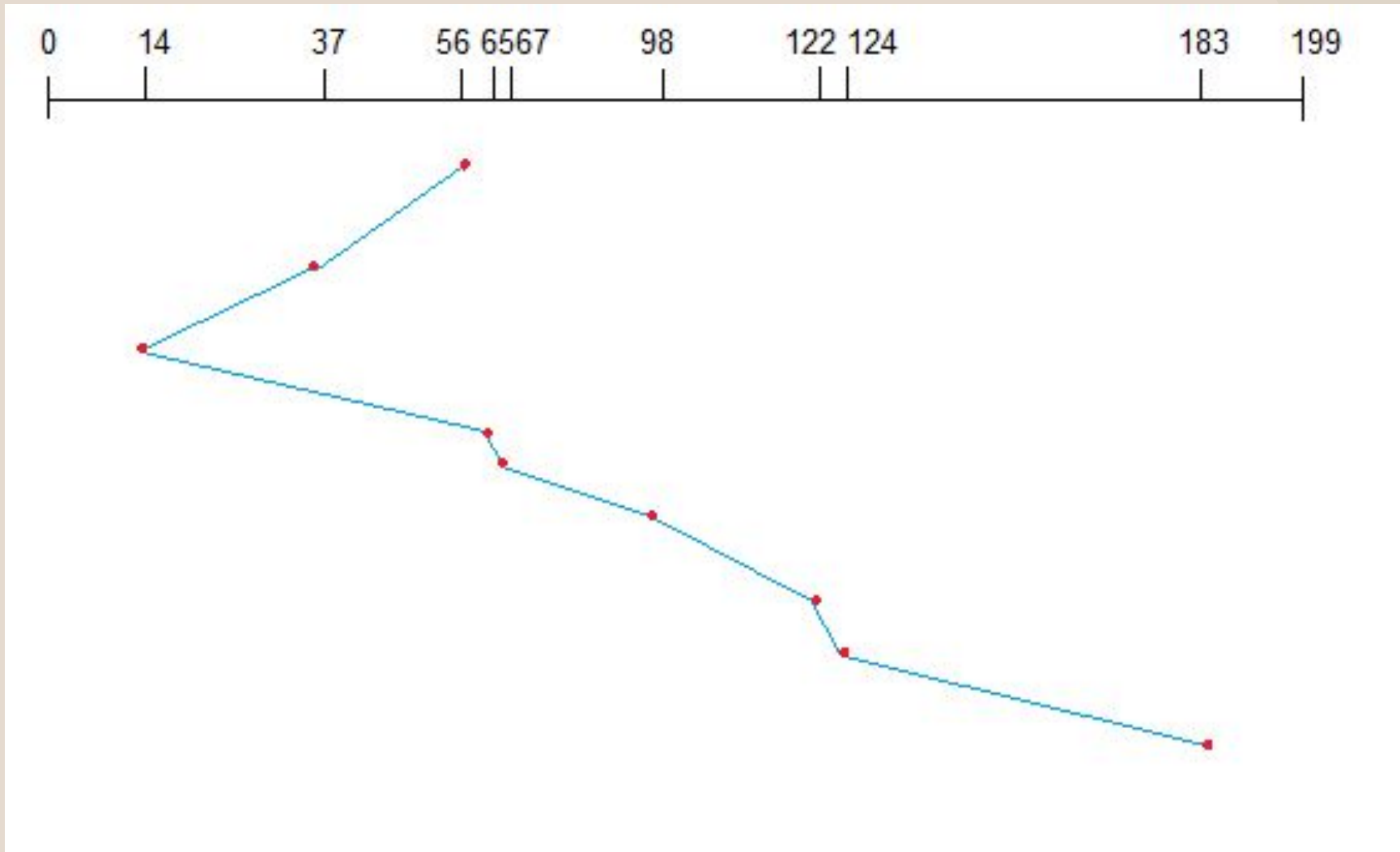
- Here the position which is closest to the current head position is chosen first. Consider the previous example where disk queue looks like,
- 98, 183, 37, 122, 14, 124, 65, 67
- Assume the head is initially at cylinder 56. The next closest cylinder to 56 is 65, and then the next nearest one is 67, then 37, 14, so on.



SCAN algorithm

SCAN algorithm

- This algorithm is also called the elevator algorithm because of its behavior. Here, first the head moves in a direction (say backward) and covers all the requests in the path. Then it moves in the opposite direction and covers the remaining requests in the path. This behavior is similar to that of an elevator. Let's take the previous example,
- 98, 183, 37, 122, 14, 124, 65, 67
- Assume the head is initially at cylinder 56. The head moves in backward direction and accesses 37 and 14. Then it goes in the opposite direction and accesses the cylinders as they come in the path.



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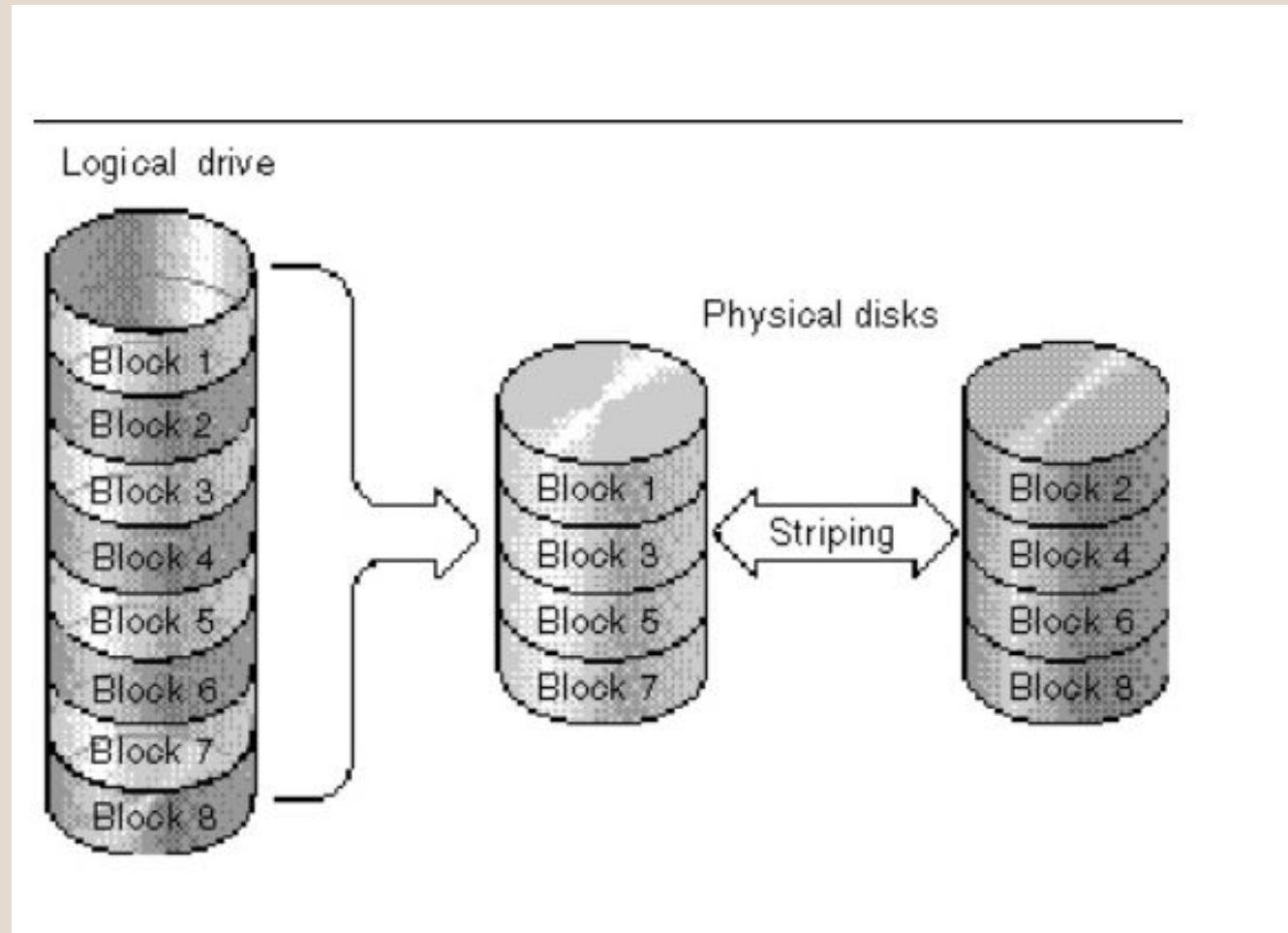
RAID(Redundant Arrays of Inexpensive Disks)

RAID is a technique that makes use of a combination of multiple disks instead of using a single disk for increased performance, data redundancy, or both.

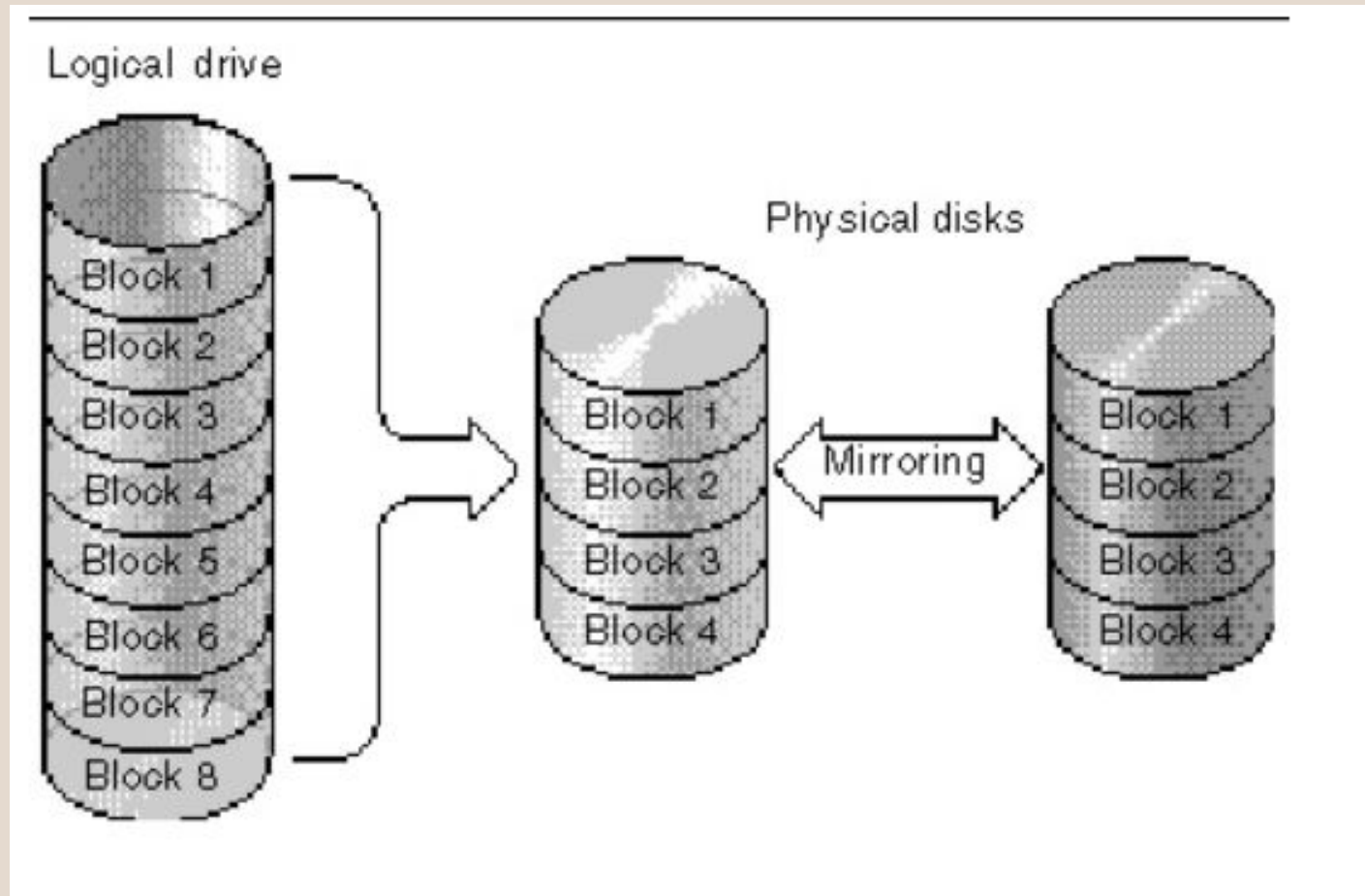
Different RAID Levels

1. RAID-0 (Stripping)
2. RAID-1 (Mirroring)
3. RAID-(1+0)
5. RAID-3 (Block-Level Stripping with Dedicated Parity)
6. RAID-5 (Block-Level Stripping with Distributed Parity)
7. RAID-6 (Block-Level Stripping with two Parity Bits)

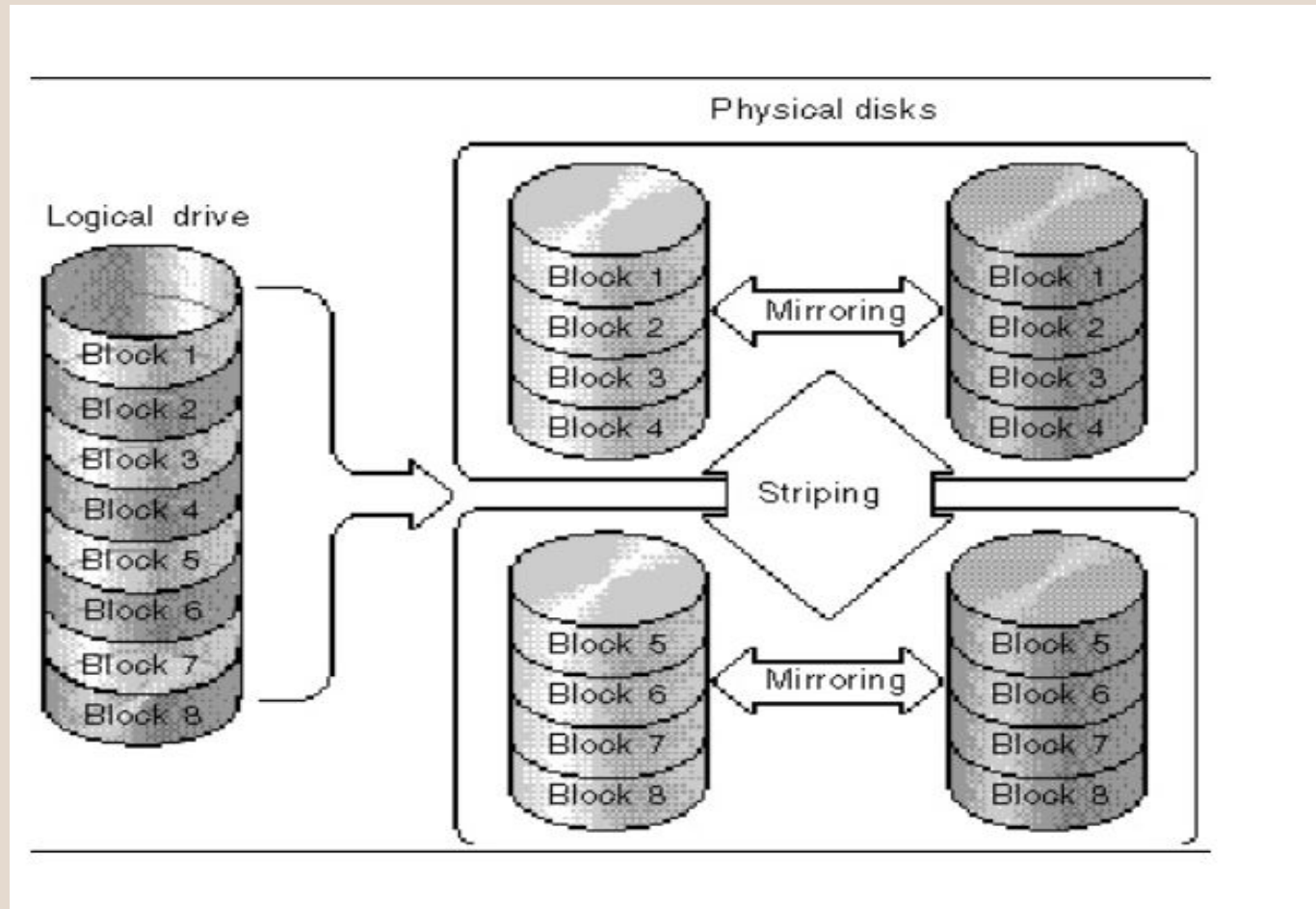
1. RAID-0 (Striping)



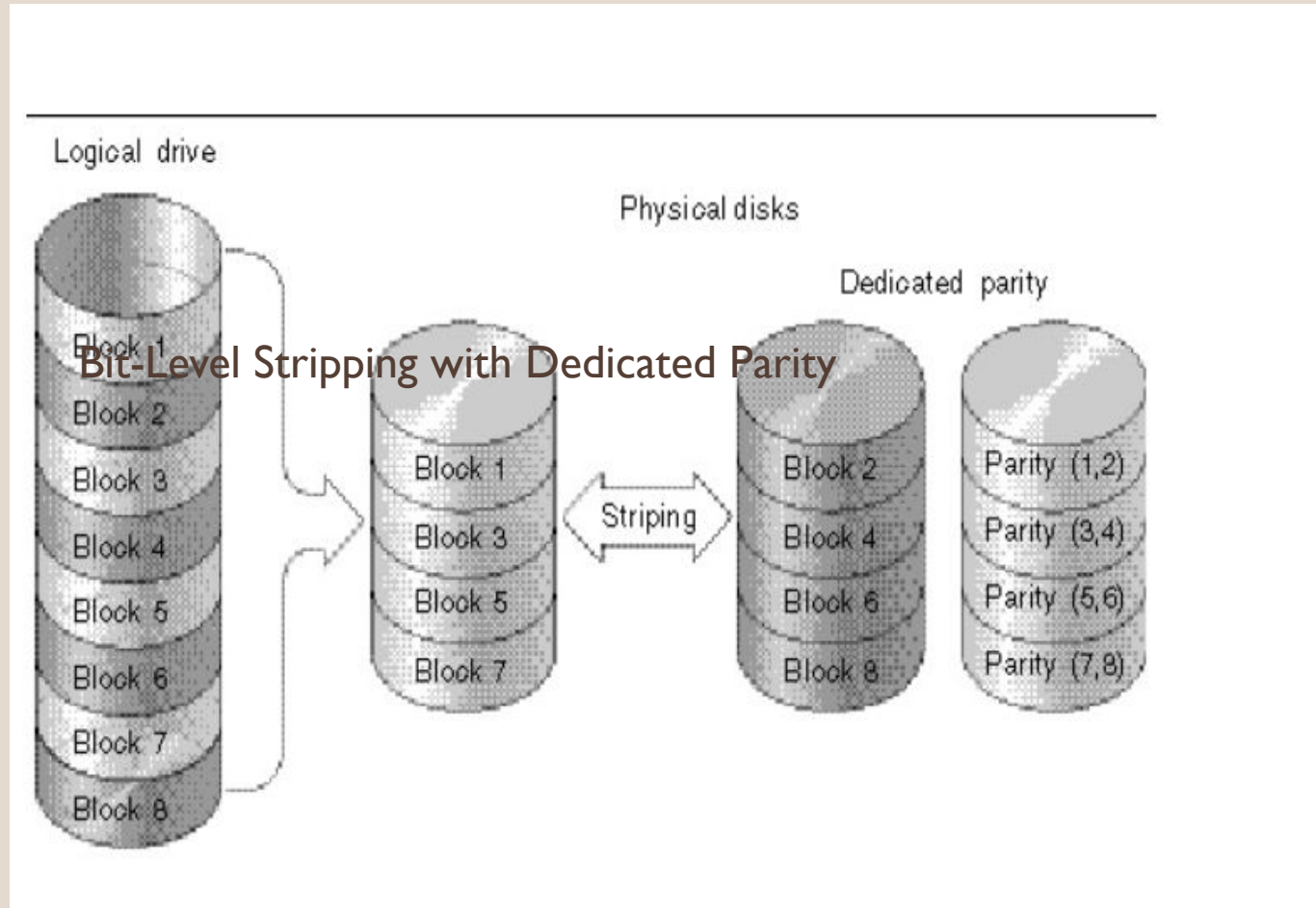
2. RAID-1 (Mirroring)



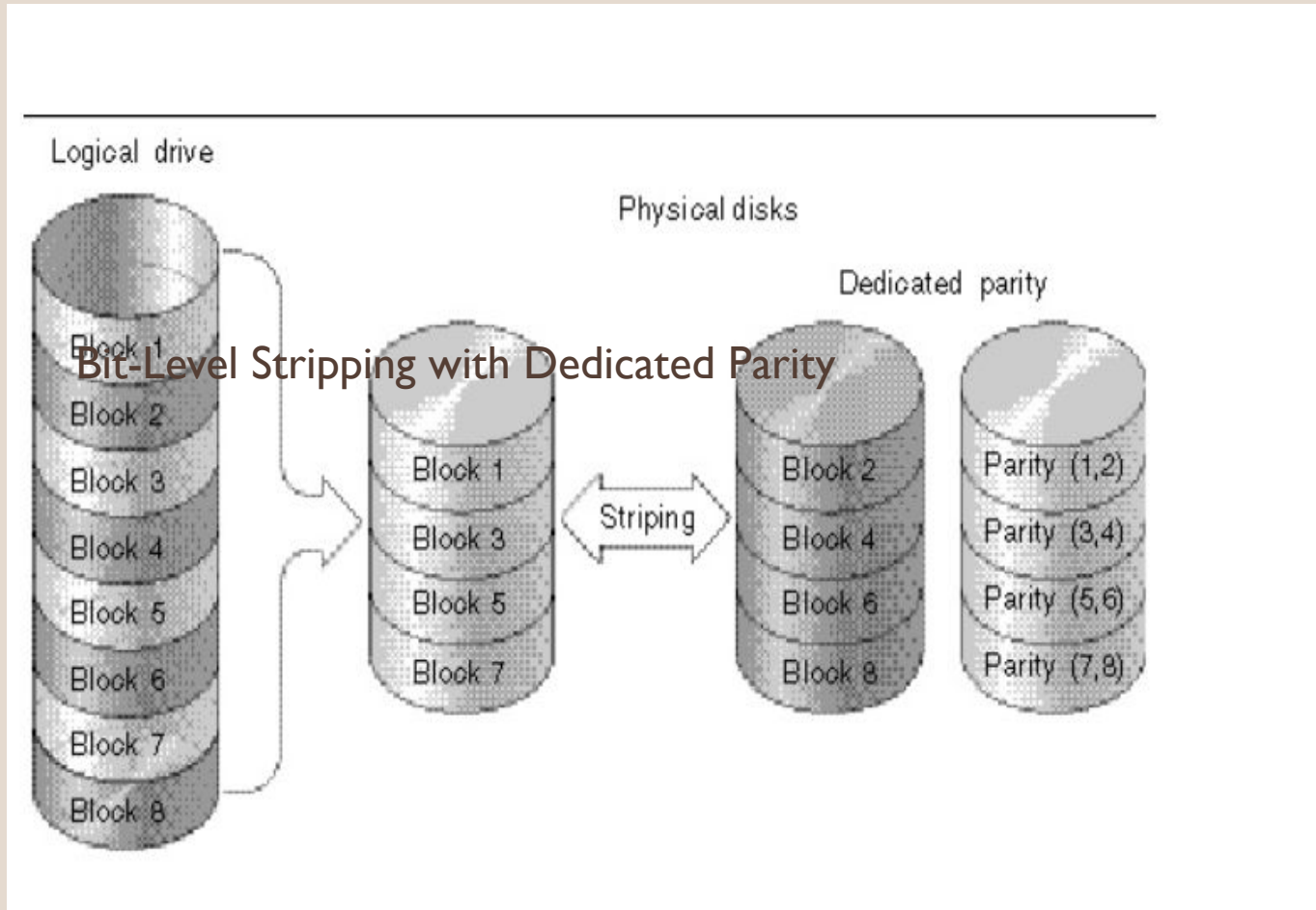
3. RAID-(1+0)



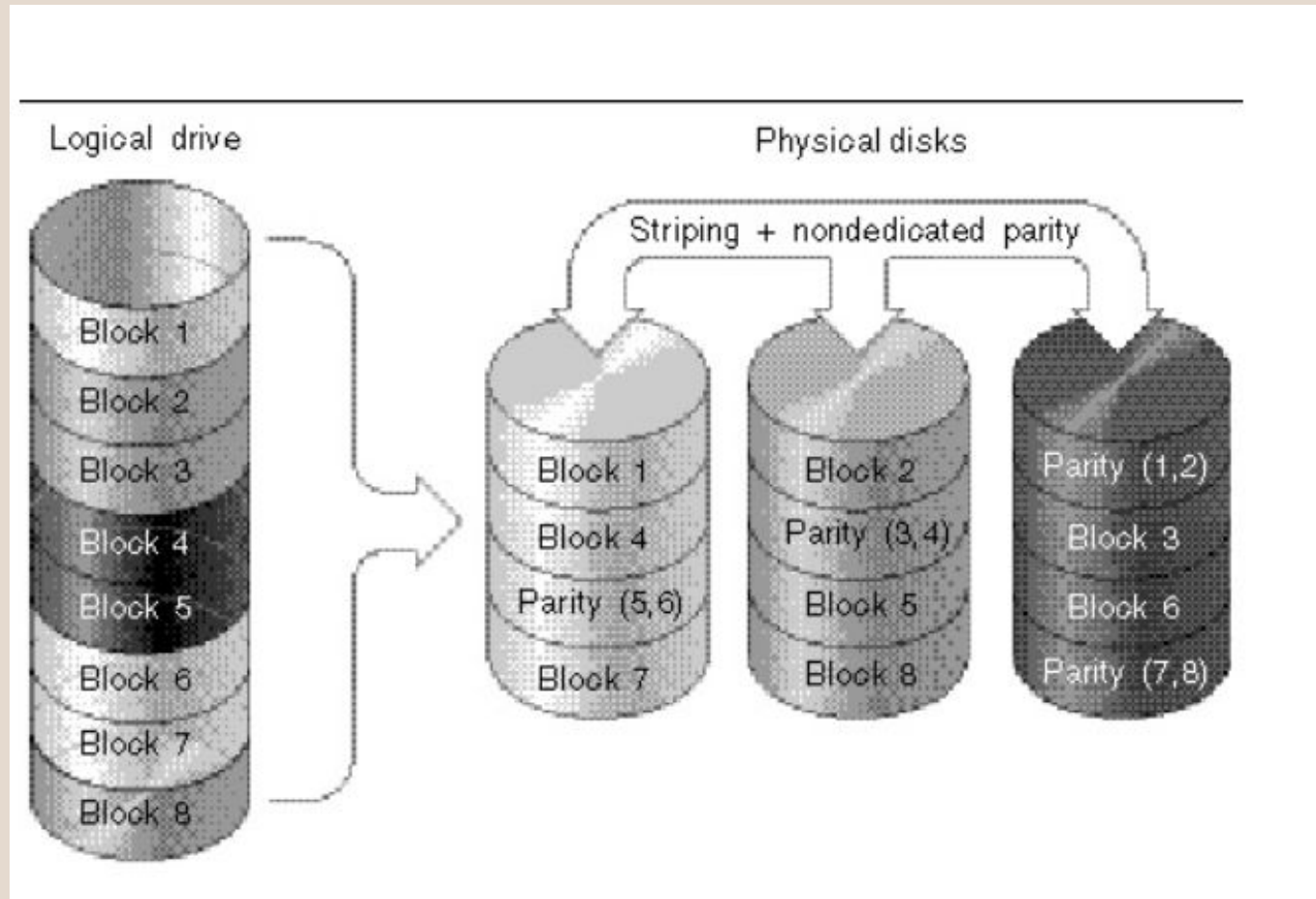
4. RAID-3(Bit-Level Stripping with Dedicated Parity)



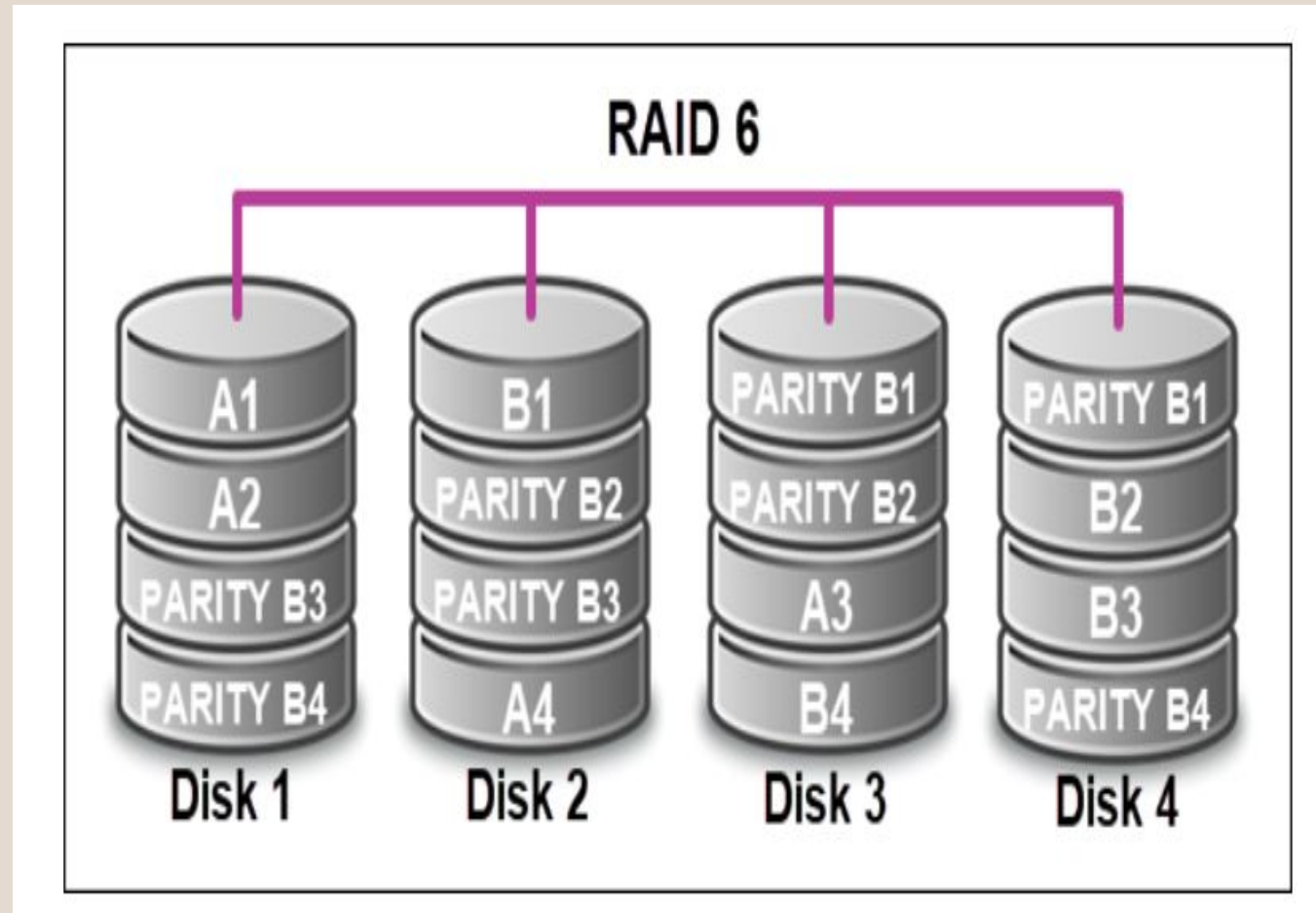
5. RAID-4(Block-Level Striping with Dedicated Parity)



6. RAID-5(Block-Level Stripping with Dedicated Parity)



6. RAID-6 (Block-Level Stripping with two Parity Bits)



The background features a light gray base with large, soft-edged organic shapes in muted red and olive green. A thin white line outlines a shape on the right. In the top left, there is a faint, light gray sketch of a leafy branch.

Thank you